

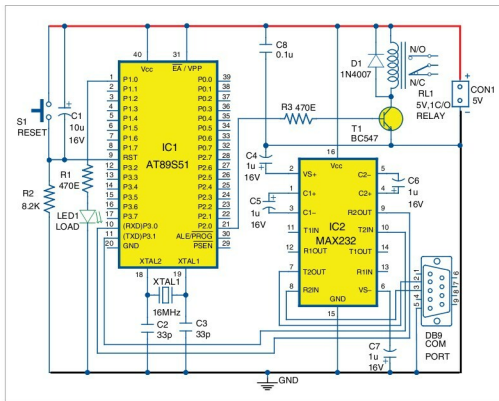


Fig. 7: Circuit diagram of the sound-operated device-control system

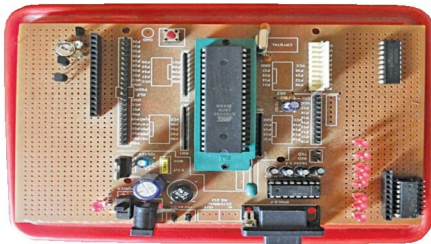


Fig. 8: Author's prototype

first-time generated wav file should be used as reference wav file or library wav file for comparison. Command should be something like 'Switch on the fan.'

Path generated in the front panel for library purpose should be given to the next VI (cross-correlation) at label named Library command. For storing command, click Run on the

block diagram of VI. The command should be at least of four seconds with at least one second between words of command. You can store or give audio command using a head-phone or directly from the inbuilt microphone of the laptop. Library command should be stored in noise-free environment.

Cross-correlation VI. The wav

file creation VI is kept in cross-correlation VI as sub-VI (VI within the VI) as shown in Fig. 3. Two different audio wav files are compared using 1D cross-correlation function block. Cross-correlation is a mathematical tool to analyse any function or continuous signal.

If both files are the same, then correlation value should be between 0 and 1 (value can be between 0.0001 to 0.67). In case of unmatched speech command, obtained value is in between 0.1E-5 and 0.9E-5. On the basis of this, threshold value is set as 0.0001. By setting very accurate threshold value, this project can be used for security and authentication purposes.

This VI will compare library file (audio command stored) and run-time command given by user. It needs a threshold value for comparison. So enter appropriate threshold value at the label named 'Threshold Value' in the front panel (Fig. 4). Threshold value should be between 0.0001 and 0.0005 using trial-and-error method.

Serial input output VI. Decision 'Y' or 'yes' (Boolean 1) is generated if output of comparison is greater than threshold value, else 'N' or 'no' (Boolean 0) is generated. 'Y' or 'N' output is transmitted to AT89S51 controller via serial communication VI (Fig. 5), which is configured as frame bits 8, baud rate 9600, parity and flow control none. Cross-correlation VI is a sub VI of serial communication VI.

Before running serial communication VI, attach USB-to-serial communication cable to the laptop. This cable is detected at the label named 'Visa Source Name' as shown in the front panel (Fig. 6). After running